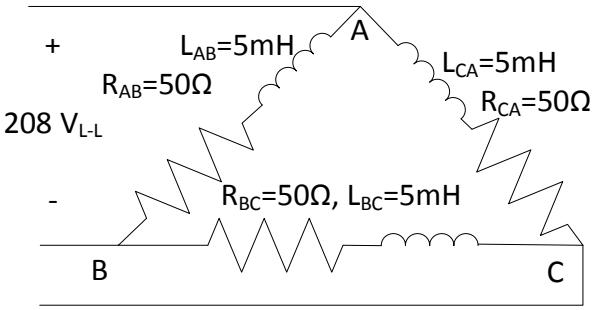
Assignment 3

1) A 300 kVA single phase transformer has a 14,400V primary and a 600V secondary. It has an impedance of 3%. At no load the secondary voltage goes as high as 605V and at full load it drops to 590V. Calculate the following:
    a. The full load primary and secondary currents
    b. The turns ratio
    c. The voltage regulation (in %)
    d. The maximum available fault current at the secondary of the transformer
2) The primary windings of three single-phase transformers are wye-connected and the secondary windings are delta-connected. In each of the single phase transformers, the primary winding has 50 turns and the secondary has 10. Three AC supplies, each 3 kV, are applied across each primary transformer. The primary line currents are 50 A. Find:
    a. The secondary line-to-line voltage of the delta-connected set for a no-load condition
    b. The phase current on the secondary side
    c. The line current of the delta-connected set on the secondary side
    d. If the secondary were wye-connected, how would (a) and (c) vary
3) A 3 phase induction motor has a synchronous speed of 1800 RPM. The motor draws 100 kW of active power. At full load its speed is 1680 RPM. Calculate the following:
    a. The number of pairs of poles per phase
    b. The slip (in %) of the motor
    c. The rated torque of the motor
    d. If the motor was operated in a country where the electrical supply frequency was 50 Hz what would the synchronous speed of the motor be?
4) For the delta-connected circuit below, calculate (assuming balanced source)
    a. the total real power, reactive power, apparent power and power factor
    b. the capacitance per phase required to bring the power factor to unity assuming a **wye**-connected three-phase capacitor bank connected in parallel
    c. the capacitance per phase required to bring the power factor to unity assuming a **delta**-connected three-phase capacitor bank connected in parallel



© K. Manson & J. Bloemink